CoSAM: Automatic Zero-Shot Co-Segmentation using DINOv2 and Segment Anything Model 2.1

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Abstract

We address the problem of automatic co-segmentation, which aims to segment common objects across multiple images without any supervision or manual prompts. Traditional methods based on low-level feature matching (e.g., SIFT) often fail under significant variations in appearance, pose, lighting, and viewpoint.

In this work, we propose **CoSAM**, a novel zero-shot framework that combines dense semantic features from DINOv2 with the powerful segmentation capability of Segment Anything Model 2.1 (SAM 2.1). DINOv2 extracts high-level semantic patch features from both images to compute a cross-image similarity heatmap. Top-K high-confidence points from this heatmap are automatically selected as prompts for SAM 2.1, which then generates accurate segmentation masks for the common object in both images.

Experimental results on various image pairs show that the proposed method successfully segments common objects in favorable conditions. However, performance degrades in cases with complex or repetitive backgrounds and large viewpoint changes. The entire pipeline requires no training, no labeled data, and no manual intervention.

1 Introduction

Co-segmentation is a fundamental task in computer vision that involves identifying and segmenting the same object(s) appearing in a set of images. It has important applications in medical imaging, autonomous driving, video editing, visual search, and instance-level image retrieval.

Traditional approaches rely on geometric or low-level feature matching techniques such as SIFT or ORB. While effective in controlled environments, these methods struggle with large variations in object appearance, scale, pose, and background clutter.

Recent advances in self-supervised vision models and foundation segmentation models open new possibilities. In this paper, we propose **CoSAM**, a fully automatic zero-shot co-segmentation system.

The key novelty of our approach lies in bridging semantic correspondence (via DINOv2) with promptable segmentation (via SAM 2.1) in a completely unsupervised manner. Unlike previous methods that require manual prompts or training on specific datasets, our method automatically generates high-quality prompts from cross-image semantic similarity, making it truly zero-shot and generalizable to unseen object categories.

2 Proposed Method

Our CoSAM framework consists of four main steps:

1. **Feature Extraction**: Both input images are resized to dimensions compatible with DINOv2's patch size (14×14). Dense patch-level features

are extracted using the pretrained DINOv2 ViT-B/14 model.

2. **Cross-Image Similarity**: Features from both images are L2-normalized. Cosine similarity is computed between every patch in Image 1 and all patches in Image 2. The maximum similarity for each patch in Image 1 is retained to form a similarity map.

3. **Heatmap & Prompt Generation**: The similarity map is reshaped into a 2D heatmap. The top-K (typically 10–15) highest-scoring locations are selected and mapped back to the original image coordinates as point prompts.

4. **Segmentation with SAM 2.1**: The generated points (with positive labels) are fed as prompts to SAM 2.1. The mask with the highest predicted IoU score is selected as the final segmentation output for both images.

2.1 Pipeline Overview

Figure 1 illustrates the complete pipeline.

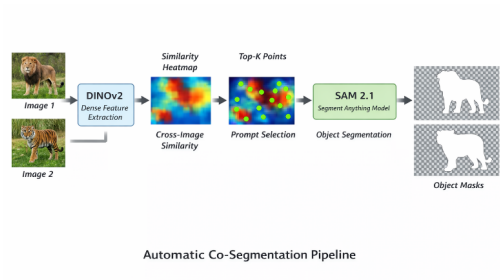

Figure 1: Overview of the proposed CoSAM pipeline. DINOv2 extracts semantic features → Cross-image similarity heatmap is generated → Top-K points are selected automatically → SAM 2.1 produces final masks.

3 Experiments and Analysis

We evaluated CoSAM on multiple pairs of images containing common objects under varying conditions (lighting, background, partial occlusion, and viewpoint differences).

3.1 Qualitative Results

Figure 2 shows representative results including the input images, similarity heatmap, and final segmentation masks.

3.2 Observations

- The method performs well when the common object is centrally located and occupies a significant portion of both images.
- Repetitive background textures (grass, roads, patterns) often dominate the similarity scores, leading to incorrect prompts.
- Partial segmentation occurs when only one part of the object (e.g., face vs. full body) receives high similarity.
- Large viewpoint or scale differences reduce the quality of the similarity heatmap.

3.3 Comparison with Traditional Methods

Table 1: Comparison of co-segmentation approaches

Method	Handles Appearance Variation	Zero-Shot	I
SIFT-based	No	Yes	
Supervised Co-Seg	Yes	No	
CoSAM (Ours)	Partially	**Yes**	

3.4 Failure Cases

Failure cases primarily occur in: - Images with multiple similar objects - Significant viewpoint or pose differences - Highly cluttered or textured backgrounds

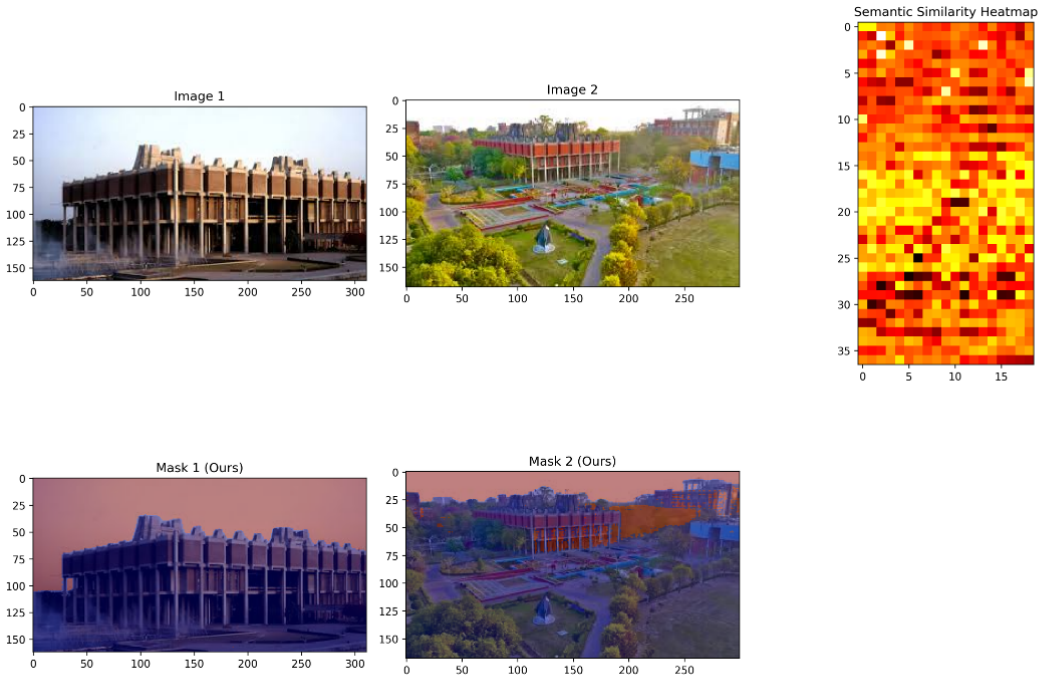

Figure 2: Qualitative results. From left to right: Image 1, Image 2, Semantic Heatmap, Mask on Image 1, Mask on Image 2. Green overlay shows the segmented common object.

4 Conclusion

We presented CoSAM, a simple yet effective zero-shot co-segmentation framework that leverages the semantic richness of DINOv2 and the precise segmentation power of SAM 2.1. The method automatically generates prompts from cross-image similarity, eliminating the need for manual intervention.

While promising in favorable conditions, the approach reveals challenges with background interference and viewpoint robustness. Future work will focus on saliency-guided prompt filtering, multi-scale similarity, and incorporating temporal consistency for video co-segmentation.

A Author Contributions

- **Nishkarsh Kaushal**: Implemented DINOv2-based dense feature extraction, cross-image similarity computation, heatmap generation, and prompt point selection logic.
- **Shaswat**: Integrated SAM 2.1 for segmentation, developed the automatic prompt generation pipeline, performed experiments, failure case analysis, and result visualization.
- Both members contributed to overall system design, report writing, result evaluation, and final documentation.

B Implementation Details

The system is implemented in PyTorch using Hugging Face Transformers. We use DINOv2 ViT-B/14 for feature extraction and SAM 2.1 (hiera-large) for segmentation. All experiments were conducted on a GPU with CUDA support.

C Complete Code

The full implementation (including the corrected notebook version) is available at: **GitHub Repository:** `https://github.com/shaswatgithub/DINOv2-SAM-2.1-Zero-Shot-Co-Segmentation`

A Streamlit web demo is also deployed where users can upload two images and view the similarity heatmap + segmented outputs in real-time. **Live App:** `https://shaswatgithub-dinov2-sam-2-1-zero-shot-co-segmentat-app1-hchqrt.streamlit.app/` **Google Colab:** `https://colab.research.google.com/drive/1zz-jxHsVtxVSbwfl8OLRBd5FhWZDr706?usp=sharing`

D Repositories and Datasets Used

- DINOv2: `https://github.com/facebookresearch/dinov2`
- SAM 2.1: `https://huggingface.co/facebook/sam2.1-hiera-large`
- SAM 2 Paper: Ravi et al., "Segment Anything in Images and Videos", arXiv:2408.00714, 2024
- DINOv2 Paper: Oquab et al., "DINOv2: Learning Robust Visual Features without Supervision", arXiv:2304.07193, 2023

Datasets used for evaluation: Publicly available internet images containing common objects (cars, animals, products, etc.). No specific benchmark dataset was used due to the zero-shot nature of the project.